Biochemistry Cume

October 20, 2006

Nucleic Acids

1. (10 points) A previous Nobel Prize was awarded to Kary Mullis for the polymerase chain reaction. Describe an experimental reason why you might want to use the polymerase chain reaction.

2. (45 points) What equipment and reagents would you need in order to carry out the polymerase chain reaction? Describe the purpose of each item in your list.

3. (10 points) Compared to protein molecules, are the tertiary structures of DNA molecules more, less or equally tolerant of changes in primary structure? Explain your answer.

4. (10 points) Describe the expected outcome of storing DNA for 18 hr in solution at pH 9.2. Explain why.

5. (10 points) Describe the expected outcome of storing RNA for 18 hr in solution at pH 9.2. Explain why.

6. (15 points) Describe briefly an experimental topic that led to a 2006 Nobel Prize in *either* Chemistry *or* Medicine. State briefly why the particular research was deemed worthy of the Nobel Prize.